

Patrick Kariuki

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EDUCATION

Johns Hopkins University

Baltimore, MD

Bachelor of Science in Computer Science

May 2026

- **Relevant Coursework:** Data Structures, Algorithms, Full-Stack JavaScript, Artificial Intelligence, Computer Systems, Object-Oriented Software Engineering, Databases, Web Security, Operating Systems, Deep Learning.

EXPERIENCE

Course Assistant

August 2024 – May 2026

Johns Hopkins University

Baltimore, MD

- Evaluated and provided technical feedback on complex algorithm and data structure implementations for 100+ students across Data Structures, Systems Programming (C/C++), and AI-Enabled Software Engineering courses.
- Led 6-person teams as technical PM for AI-Enabled Software Engineering, advising on LLM integration, prompt engineering, RAG pipelines, and deployment architecture, shipping 3 full-stack AI applications.

Tech Developer Intern

May 2024 – August 2024

Sponsors for Educational Opportunity (SEO)

New York, NY

- Built StarSight, a full-stack stargazing prediction platform integrating real-time weather data, light pollution metrics, and astronomical databases to provide location-based visibility forecasts for 50 beta users across 3 states.
- Engineered Flask REST API with secure user authentication and PostgreSQL database; implemented Redis caching layer reducing API response time from 800ms to 200ms for repeated location queries.
- Trained constellation recognition model using Principal Component Analysis for feature extraction on 10,000 labeled astronomical images, achieving 85% accuracy; deployed via Docker with Ansible-managed CI/CD pipeline.

PROJECTS

Applications Tracker | *React, Convex, TanStack, Vercel AI SDK*

June 2026 – August 2026

- Engineered a full-stack application with an AI email-parsing pipeline, using scheduled Convex functions, that reads a user's inbox, classifies application emails, and auto-updates statuses on a real-time reactive backend, with a human-in-the-loop review queue for low-certainty cases.

SDB | *C++, Docker*

March 2026 – May 2026

- Built an x86-64 debugger in C++ supporting breakpoints, stack unwinding, and source-level stepping via ptrace syscalls and DWARF debug info parsing.

2D Game Engine | *C++, Raylib, OpenGL, Lua*

January 2026 – May 2026

- Architected a 2D game engine in C++ from scratch with an entity-component-system, broadphase collision detection (sort-and-sweep), layered rendering via OpenGL, and custom memory allocators; shipped a complete tower defense game as the capstone demo.

GoCI | *Go, Git*

January 2026 – February 2026

- Built a CLI continuous integration tool in Go automating build, test, and lint pipelines with git hook integration; used as the deployment workflow for all personal projects.

LEADERSHIP

President, Software Engineering Club

May 2025 – May 2026

- Secured \$500 in corporate sponsorship and organized 5+ technical workshops on career prep, interview skills, and AI-enabled software engineering for 75+ student members.
- Founded HopBuilds, a project-based learning initiative where students build real applications for academic credit; led development of Hopkins Lost and Found platform deployed campus-wide.
- Grew active membership 40% through hands-on coding events and industry networking sessions.

TECHNICAL SKILLS

Programming: Python, C/C++, Java, JavaScript/TypeScript, Go, HTML/CSS

Frameworks: React, Node.js, Flask, Django, FastAPI, NumPy, scikit-learn, PyTorch

Databases: PostgreSQL, MongoDB, Firebase, SQLite, DynamoDB

Developer Tools: Git, Docker, Ansible, Cursor